

NAME: _____

- Each question is worth 5 points, 20 questions total and 1 bonus
- You should have 13 pages total
- You have the whole class for your attempt
- You should not have anything on you aside from your pen/ pencil and TI-30Xa calculator
- Do not use a red pen
- If you run out of space, you may continue your work on the blank reverse side or the last blank page as long as your work is clearly indicated
- Cancel working you do not want considered in your final answer. Show all working relevant to your final answer. Simplify and box all final answers
- If a numerical answer is not exact, express it using fractions/ radicals/ etc.

Q.1. Simplify the following expression:

$$\left(\frac{a^2 \cdot a^5}{b^{-2}}\right)^2 \cdot b^4$$

Combining like powers inside the bracket (using the product rule):

$$\begin{aligned} &= \left(\frac{a^{2+5}}{b^{-2}}\right)^2 \cdot b^4 \\ &= \left(\frac{a^7}{b^{-2}}\right)^2 \cdot b^4 \end{aligned}$$

Next we distribute the power over each term in the quotient and then apply the power rule to simplify:

$$\begin{aligned} &= \frac{(a^7)^2}{(b^{-2})^2} \cdot b^4 \\ &= \frac{a^{7 \cdot 2}}{b^{(-2) \cdot 2}} \cdot b^4 \\ &= \frac{a^{14}}{b^{-4}} \cdot b^4 \end{aligned}$$

Lastly, we combine like powers of b via the quotient rule:

$$= a^{14} \cdot b^{4-(-4)} = a^{14} \cdot b^8$$

There are no more operations left to do or like powers to combine, so our final answer is $a^{14} \cdot b^8$.

Q.2. Simplify the following expression where a, b may be any real numbers:

$$\sqrt{a^3 \cdot b^2} \cdot \sqrt{a^7 \cdot b^4}$$

We start by combining the two radicals into a single radical and multiplying like powers together via the product rule:

$$\begin{aligned}
&= \sqrt{a^3 \cdot b^2 \cdot a^7 \cdot b^4} \\
&= \sqrt{a^{3+7} \cdot b^{2+4}} \\
&= \sqrt{a^{10} \cdot b^6}
\end{aligned}$$

There are a few ways to simplify this radical. One way, since we are taking a square root, is to think about how many "pairs" of a and b may be formed inside the radical and how many copies are left over. We can pull one copy of a and b out of the radical for each corresponding pair inside the radical. We have a^{10} inside the radical, i.e. 5 pairs of a inside the radical with no copies of a left over. Similarly, we have 3 pairs of b inside the radical with no copies of b left over.

So our answer is $a^5 \cdot b^3$, and we're done...except! The problem says a and b can be **any** real number. If a and b are any real number, $a^5 \cdot b^3$ could very well be negative (e.g. $a = 1, b = -1$), but this is a result of a principal root which can never be negative. We need to adjust our answer by adding an absolute value that covers both variables, and there are many valid ways to write this final answer, e.g. $|a^5 \cdot b^3|$, $|a^5| \cdot |b^3|$, $|a|^5 \cdot |b|^3$, etc. Remember the big exception we pointed out earlier in the course about powers and radicals cancelling: in general $\sqrt{x^2}$ is not simply x , it's $|x|$. This is true actually for all even powers. This fact is another, more concrete way to think about where the absolute value in our solution comes from.

Q.3. Simplify and rationalize the denominator:

$$\frac{x^2 \cdot y}{\sqrt[3]{y^5}}$$

There are a few ways to complete this problem. One is to start by simplifying the denominator radical (again thinking about how many "triples" of y can be formed and how many copies of y will be left over):

$$\frac{x^2 \cdot y}{\sqrt[3]{y^5}} = \frac{x^2 \cdot y}{y \cdot \sqrt[3]{y^2}} = \frac{x^2}{\sqrt[3]{y^2}}$$

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and now we are able to use our formula for rationalizing denominators $\frac{a}{\sqrt[n]{b^k}} = \frac{a \cdot \sqrt[n]{b^{n-k}}}{b}$ (this correctly rationalizes your expression when $k < n$). In this problem b is y , $n = 3$, and $k = 2$, so:

$$\frac{x^2}{\sqrt[3]{y^2}} = \frac{x^2 \cdot \sqrt[3]{y}}{y}$$

Put more simply, we simply multiplied the above fraction's numerator and denominator by $\sqrt[3]{y}$, allowing us to get a full triple in the denominator and simplify that radical completely. So our final answer

is $\boxed{\frac{x^2 \cdot \sqrt[3]{y}}{y}}$.

Q.4. Rationalize the numerator:

$$\frac{5 + \sqrt{x}}{5 - \sqrt{x}}$$

Since we are rationalizing a numerator containing a square-root summed with another number, we ought to multiply and divide by the conjugate of the numerator, i.e. $5 - \sqrt{x}$

$$\begin{aligned} &= \frac{5 + \sqrt{x}}{5 - \sqrt{x}} \cdot \frac{5 - \sqrt{x}}{5 - \sqrt{x}} \\ &= \frac{(5 + \sqrt{x}) \cdot (5 - \sqrt{x})}{(5 - \sqrt{x}) \cdot (5 - \sqrt{x})} \\ &= \frac{5^2 - (\sqrt{x})^2}{5^2 - 2 \cdot 5 \cdot \sqrt{x} + (\sqrt{x})^2} \\ &= \frac{25 - x}{25 - 10\sqrt{x} + x} \end{aligned}$$

so our final answer is $\boxed{\frac{25 - x}{25 - 10\sqrt{x} + x}}$.

Q.5. Simplify the following difference of rational expressions and **fully factor** the numerator and denominator in your final answer:

$$\frac{3}{x-2} - \frac{6}{x^2+x-6}$$

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First, we determine the LCD (least common denominator). We need to fully factor each denominator to figure this out. $x - 2$, being a linear polynomial (whose coefficient GCD is 1) is fully factored as is. $x^2 + x - 6$, however, is quadratic. We can factor this further via the p and q method. For this quadratic polynomial, $a = 1, b = 1, c = -6$, so we are looking for p and q such that $p \cdot q = -6$ and $p + q = 1$. It is easy to see that $p = 3, q = -2$ works here.

Now we split the middle term and factor by grouping:

$$\begin{aligned} x^2 + x - 6 &= x^2 + 3x - 2x - 6 \\ &= x(x + 3) - 2(x + 3) \\ &= (x - 2)(x + 3) \end{aligned}$$

so $x^2 + x - 6$ fully factors as $(x - 2)(x + 3)$. It follows that our LCD is $x^2 + x - 6 = (x - 2)(x + 3)$ itself. The second rational expression already has this denominator as its LCD and is fine as is. The first rational expression has a different denominator - its denominator only has the $x - 2$ factor and is missing the $x + 3$ factor.

Rewriting the first rational expression:

$$\frac{3}{x - 2} = \frac{3}{x - 2} \cdot \frac{x + 3}{x + 3} = \frac{3(x + 3)}{(x - 2)(x + 3)} = \frac{3x + 9}{(x - 2)(x + 3)}$$

Finally, adding the two fractions together (and factoring the final result):

$$\begin{aligned} \frac{3}{x - 2} - \frac{6}{x^2 + x - 6} &= \frac{3x + 9}{(x - 2)(x + 3)} - \frac{6}{(x - 2)(x + 3)} \\ &= \frac{3x + 9 - 6}{(x - 2)(x + 3)} \\ &= \frac{3x + 3}{(x - 2)(x + 3)} \\ &= \frac{3(x + 1)}{(x - 2)(x + 3)} \end{aligned}$$

so our final answer is $\frac{3(x+1)}{(x-2)(x+3)}$.

Q.6. Simplify the following quotient of rational expressions and **fully factor** the numerator and denominator in your final answer:

$$\frac{-2x^2 + 14x - 20}{x - 2} \div \frac{x^2 - 25}{3x}$$

Again, we start by factoring every numerator and denominator polynomial fully. $3x$ and $x - 2$ are fully factored as is. $-2x^2 + 14x - 20$ has a coefficient GCD of -2 . Factoring this out, the polynomial equals $-2(x^2 - 7x + 10)$. Applying p and q method to the polynomial $x^2 - 7x + 10$, we see that it may be factored as $(x - 5)(x - 2)$, so $-2x^2 + 14x - 20 = -2(x - 5)(x - 2)$.

And $x^2 - 25$ may be factored via the difference of squares identity as $(x - 5)(x + 5)$.

Rewriting our quotient:

$$\begin{aligned} \frac{-2x^2 + 14x - 20}{x - 2} \div \frac{x^2 - 25}{3x} &= \frac{-2(x - 5)(\cancel{x - 2})}{\cancel{x - 2}} \div \frac{(x - 5)(x + 5)}{3x} \\ &= -2(\cancel{x - 5}) \cdot \frac{3x}{(\cancel{x - 5})(x + 5)} \\ &= \frac{-6x}{x + 5} \end{aligned}$$

so our final answer is $\frac{-6x}{x + 5}$.

Q.7. Find all real solutions of the following equation:

$$(2x - 2)^2 = (4x + 2) \cdot (x - 4)$$

We begin by expanding both sides of the equation:

$$\begin{aligned}(2x - 2)^2 &= (4x + 2) \cdot (x - 4) \\ (2x)^2 + 2 \cdot (2x) \cdot (-2) + (-2)^2 &= 4x \cdot x + 4x \cdot (-4) + 2 \cdot x + 2 \cdot (-4) \\ \cancel{4x^2} - 8x + 4 &= \cancel{4x^2} - 14x - 8 \\ 6x &= -12 \\ x &= -2\end{aligned}$$

we see that after expansion, the x^2 terms may be eliminated and we simply have a linear equation with a final solution of $x = -2$.

Q.8. 550 people attended the premiere of a film. Each adult ticket cost \$10 and each child ticket cost \$5. If box office revenue totalled \$5,000, how many adults and how many children attended the premiere? (**Hint:** you will need two variables and two equations)

We introduce one variable for each of our unknowns. Let x be the number of adults who attended the premiere and y be the number of children who attended the premiere.

The total number of people who attended the premiere is 550, so one equation is:

$$x + y = 550 \tag{1}$$

The second equation is trickier. This second equation will be about quantifying total revenue. We know that the total revenue was \$5,000. At the same time, since each adult ticket cost \$10, the total revenue from just adult tickets should be $10x$. Similarly the total revenue from just child tickets should be $5y$. Altogether this revenue is $10x + 5y$ and this must equal the total of 5,000:

$$10x + 5y = 5,000$$

which dividing through by 5 may be simplified to:

$$2x + y = 1,000 \quad (2)$$

We now need to combine these two equations to solve for one variable by substituting the other. For instance, the first equation tells us that $y = 550 - x$. Substituting this into the second equation gives us:

$$2x + y = 1,000 \quad (3)$$

$$\Rightarrow 2x + (550 - x) = 1,000 \quad (4)$$

$$\Rightarrow x + 550 = 1,000 \quad (5)$$

$$\Rightarrow x = 450 \quad (6)$$

Now that we know what x is, we can plug this value back into either equation to solve for y . For instance, plugging it back into the first equation:

$$x + y = 550 \quad (7)$$

$$\Rightarrow 450 + y = 550 \quad (8)$$

$$\Rightarrow y = 100 \quad (9)$$

so 450 adults and 100 children attended the premiere.

Q.9. Simplify the following quotient of complex numbers into a single complex number of the form $a + bi$ with a, b real:

$$\frac{4 - 3i}{1 - 2i}$$

The denominator is $1 - 2i$. This value has conjugate $1 + 2i$. Remember that when we multiply a complex number by its complex conjugate, we get its "length squared", i.e. $(a + bi) \cdot (a - bi) = a^2 + b^2$. Remember also that $i^2 = -1$. We need to multiply and divide the fraction by this conjugate value $1 + 2i$, and then simplify:

$$\begin{aligned}
\frac{4-3i}{1-2i} &= \frac{4-3i}{1-2i} \cdot \frac{1+2i}{1+2i} \\
&= \frac{(4-3i) \cdot (1+2i)}{(1-2i) \cdot (1+2i)} \\
&= \frac{4 \cdot 1 + 4 \cdot (2i) - 3i \cdot 1 - 3i \cdot (2i)}{1^2 + 2^2} \\
&= \frac{4 + 8i - 3i + 6}{5} \\
&= \frac{10 + 5i}{5} \\
&= 2 + i
\end{aligned}$$

so our final answer is $2 + i$.

Q.10. State the quadratic formula for the general equation $ax^2 + bx + c = 0$. Then use this formula to find all **complex** solutions of the equation $-4x^2 + 8x - 5 = 0$:

The general quadratic formula for the quadratic equation $ax^2 + bx + c = 0$ is:

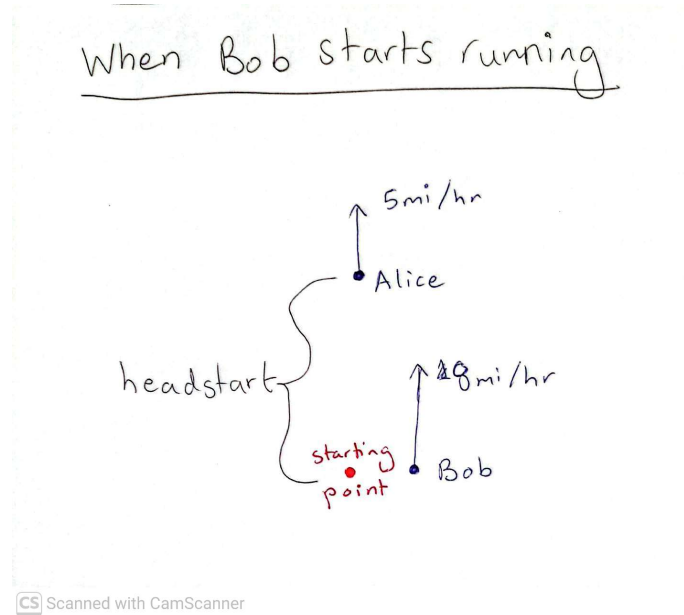
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

In this problem, $a = -4$, $b = 8$, $c = -5$, so:

$$\begin{aligned}
x &= \frac{-8 \pm \sqrt{8^2 - 4(-4)(-5)}}{2(-4)} \\
&= \frac{-8 \pm \sqrt{64 - 80}}{-8} \\
&= \frac{-8 \pm \sqrt{-16}}{-8} \\
&= \frac{-8 \pm 4i}{-8} \\
&= \frac{2 \pm i}{2}
\end{aligned}$$

where in the last step, we cancelled a -4 from the numerator and denominator. So our final answer is $\frac{2 \pm i}{2} = 1 \pm \frac{i}{2}$.

Q.11. Alice starts at the beginning of a runner's path and runs at a constant rate of 5 mi/hr. 20 minutes later Bob starts running from the same starting point at a constant rate of 8 mi/hr. How long **in minutes** will it take for Bob to catch up to Alice? (it helps to draw a diagram on the reverse side of the page!)



Alice has a headstart over Bob. She started running at the same starting point as Bob, but 20 minutes earlier than he did. How much of a headstart did that 20 minutes give her? Well she is running at a constant rate of 5 mi/hr throughout this whole problem, so during that 20 minutes she covered:

$$(5 \text{ mi/hr}) \cdot (20 \text{ mins}) = (5 \text{ mi/hr}) \cdot \left(\frac{1}{3} \text{ hr}\right) = \frac{5}{3} \text{ mi}$$

meaning that Alice has a $\frac{5}{3}$ mi headstart over Bob. So given this headstart and their constant speeds of 5 mi/hr and 8 mi/hr, respectively, at time t hours after Bob has started running, Alice is $\frac{5}{3} + 5t$ miles from the starting point and Bob is $8t$ miles from the starting point. Bob catches up to Alice when these distances become equal. Setting them equal and solving for t :

$$\begin{aligned}\frac{5}{3} + 5t &= 8t \\ \Rightarrow \frac{5}{3} &= 3t \\ \Rightarrow t &= \frac{5}{9}\end{aligned}$$

so Bob catches up to Alice $\frac{5}{9}$ hr ≈ 0.56 hr or $\frac{100}{3}$ mins ≈ 33 mins after he starts running.

Q.12. Find all real solutions of the following equation using the zero-factor theorem:

$$2x^3 - 3x^2 - 8x + 12 = 0$$

This is a four-term polynomial, so it makes sense to try factoring by grouping:

$$\begin{aligned}2x^3 - 3x^2 - 8x + 12 &= 0 \\ \Rightarrow x^2(2x - 3) - 4(2x - 3) &= 0 \\ \Rightarrow (x^2 - 4)(2x - 3) &= 0\end{aligned}$$

But we are not done yet, since the factor $x^2 - 4$ is a difference-of-squares and can be factored further. $x^2 - 4 = (x - 2)(x + 2)$, so our equation with the left side now fully factored is:

$$(x - 2)(x + 2)(2x - 3) = 0$$

By the zero-factor theorem, the only way the left side can equal 0 is if at least one of the factors on the left side is 0, i.e.

$$\begin{array}{lll}x - 2 = 0 & \text{OR } x + 2 = 0 & \text{OR } 2x - 3 = 0 \\ x = 2 & \text{OR } x = -2 & \text{OR } x = \frac{3}{2}\end{array}$$

So our final answer is $x = 2, -2, \frac{3}{2}$.

Q.13. Dina is taking a class in which her total grade is determined only by her scores on 3 midterm tests and 1 final exam. The final exam is worth half her total grade and the midterm tests all have equal weight. On her midterm tests, Dina scored 94%, 83%, and 87%, respectively. What percentage does Dina need to get on the final exam to achieve a total grade of exactly 90%?

We let x represent this unknown percentage she needs to get on her final exam to achieve a total grade of %90.

Since her final exam comprises half of her total grade, her midterm scores must comprise the remaining half of her total grade.

Dina's average score on her midterm tests is $= \frac{94+83+87}{3} = 88$. This average midterm score of 88 and her unknown final exam score of x are both equally weighted (since the final exam and the midterm scores each comprise half the total grade). So for Dina to get a total grade of %90, we must have:

$$\begin{aligned}\frac{88 + x}{2} &= 90 \\ 88 + x &= 180 \\ x &= 92\end{aligned}$$

So Dina needs a score of %92 on the final exam to get a total grade of exactly %90.

Q.14. Find all **real** solutions of the equation:

$$x^4 - x^2 - 12 = 0$$

This is "quadratic-type" equation, since (ignoring the constant term) the powers of x we see are in a 2 : 1 ratio. Specifically, $x^4 = x^{2 \cdot 2} = (x^2)^2$. If we introduce a substitution, $u = x^2$, we can then transform the equation from a 4th degree equation in x into a quadratic equation in u :

$$x^4 - x^2 - 12 = 0$$

$$u^2 - u - 12 = 0$$

This quadratic equation may be solved in many ways. For instance, the left side may be factored easily via the p and q method:

$$u^2 - u - 12 = 0$$

$$(u - 4)(u + 3) = 0$$

and so by the zero-factor theorem, either $u = 4$ or $u = -3$. We are not done yet, since we have only found the possible values of u , but not the possible values of x . Since $u = x^2$, $x = \pm\sqrt{u}$. If $u = 4$, we get that $x = \pm\sqrt{4} = \pm 2$, which are real solutions. But if $u = -3$, we get that $x = \pm\sqrt{-3} = \pm i\sqrt{3}$, which are non-real solutions. So the only **real** solutions to the equation $x^4 - x^2 - 12 = 0$ are $x = \pm 2$.

Q.15. Find all real solutions of the following equation. Make sure to **plug them back in to the equation** and check they are correct:

$$x - 1 = -\sqrt{7 - x}$$

We want to rearrange and solve for x somehow, but there is a pesky square-root in the way. Let's get rid of it by squaring by both sides first and seeing what we get:

$$(x - 1)^2 = (-\sqrt{7 - x})^2$$

$$(x - 1)^2 = 7 - x$$

$$x^2 - 2x + 1 = 7 - x$$

$$x^2 - x - 6 = 0$$

This is a quadratic equation, which again may be solved in a number of ways (we may factor the left side as $(x - 3)(x + 2)$, for instance) and it has solutions $x = 3, -2$.

But do these solutions actually work or are they extraneous? Let's plug them back into the original equation. For $x = 3$

$$\begin{aligned} 3 - 1 &\stackrel{?}{=} -\sqrt{7 - 3} \\ 2 &\neq -2 \end{aligned}$$

so $x = 3$ is actually **NOT** a true solution and is only extraneous.
How about $x = -2$? :

$$\begin{aligned} -2 - 1 &\stackrel{?}{=} -\sqrt{7 - (-2)} \\ -3 &= -3 \end{aligned}$$

and that checks out! So our final answer is $x = -2$ only.

Q.16. Find all real solutions of the equation:

$$\sqrt[3]{x^2 - 6x} = -2$$

There is again a very annoying root obstructing progress. This time it's a cube-root. Let us cube both sides and get rid of it:

$$\begin{aligned} \left(\sqrt[3]{x^2 - 6x}\right)^3 &= (-2)^3 \\ \Rightarrow x^2 - 6x &= -8 \\ \Rightarrow x^2 - 6x + 8 &= 0 \end{aligned}$$

Another quadratic equation that may be solved in a number of ways (we may factor the left side as $(x - 4)(x - 2)$) and it has solutions $x = 2, 4$. No extra verification is needed in this problem, i.e. you did not need to plug these values back in to the original equation and check that they made sense (they both work). That verification is really only needed with even-order root equations, not with odd-order root equations (at least when we are only interested in real solutions).

Q.17. The equation for the total amount accrued via simple interest is $A = P + Prt$ where P denotes the principal. Use this equation to express P in terms of the other variables, A, r, t (total amount, rate of interest, and time, respectively).

We see a couple of P -terms on the right side. But how can we get rid of the other variables on that side of the equation and fully isolate P ?

It helps if we can somehow reduce the number of copies of P present first (the more copies of a variable present, the harder it is to isolate for said variable). Can we combine the two P terms somehow? Well the first P term has no explicit coefficient, but we can take that to mean an "invisible 1" coefficient. The second P term has a coefficient of rt . Factoring out the P :

$$A = P(1 + rt)$$

Now with only one copy of P , we can freely rearrange and solve for P in terms of the other variables. Dividing both sides by $1 + rt$:

$$\frac{A}{1 + rt} = P$$

So we have $P = \frac{A}{1 + rt}$ as an expression of P in terms of the other variables.

Q.18. Find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$-5 \leq -3x + 1 < 10$$

We need to isolate for x . We just have to be careful about two things. First, whatever operation we perform on any of the three expressions above, we must perform on the other two expressions as well (just like an equation - whatever operation you do on one side, you must also do on the other side). Second, if we multiply/ divide by a negative number, we must flip all inequality signs. With all this in mind:

$$\begin{aligned} -5 &\leq -3x + 1 < 10 \\ \Rightarrow -6 &\leq -3x < 9 \\ \Rightarrow 2 &\geq x > -3 \end{aligned}$$

We reversed the inequality signs in the last step, because we divided by -3 . Written more intuitively, we have $-3 < x \leq 2$, so our solution in interval notation is $(-3, 2]$.

Q.19. State the **domain** of this inequality. Then find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$\frac{4}{|x+1|} > 1$$

First, let us state the domain of this inequality. The only domain restriction here comes from the division on the left side. Every other operation is valid for every real number/ pair of real numbers, except for division in which the denominator must not be 0. So $|x+1| \neq 0$, i.e. our domain is $x+1 \neq 0 \Rightarrow x \neq -1$ (in interval notation our domain is $(-\infty, -1) \cup (-1, \infty)$).

Now, let's try and solve the inequality. Ordinarily, it would be a bad idea to try and get rid of a denominator like $x+1$ from an inequality by just multiplying both sides by $x+1$, since $x+1$ could be either positive or negative and so we won't know which way to flip the sign. But here we don't have just $x+1$, we have $|x+1|$. $|x+1|$ will be positive as long as $x \neq -1$ (which is our domain anyway), so we may multiply by both sides by $|x+1|$ for any x in our domain without affecting the inequality sign:

$$\begin{aligned} \frac{4}{|x+1|} &> 1 \\ 4 &> |x+1| \\ |x+1| &< 4 \end{aligned}$$

Remember that this can be interpreted as saying $x+1$ is distance less than 4 from the origin of the real line (i.e. from 0). This means:

$$\begin{aligned} -4 &< x+1 < 4 \\ \Rightarrow -5 &< x < 3 \end{aligned}$$

So it looks like our solution is just the interval $(-5, 3)$, right? But wait! We need to remember our domain restriction. We know that we cannot admit the value $x = -1$. Our interval $(-5, 3)$ combined with the restriction $x \neq -1$ gives us our final answer $(-5, -1) \cup (-1, 3)$.

Q.20. Find all real solutions to the inequality and write your answer in interval notation (draw a sketch of your solution set if confused):

$$\frac{x(x-2)}{x+3} < 0$$

We need to interpret this inequality as a statement about "signs". This inequality is saying that the rational expression on the left side should be negative. We need to build a sign-chart/ sign-table to figure out which interval(s) this rational expression is negative on.

Step 1): find all the roots of every factor in the rational expression. Our factors here are clearly written out for us: x , $x - 2$, and $x + 3$. They have roots 0, 2, and -3 , respectively. Written in ascending order, the roots are $-3, 0, 2$.

Step 2): Now we need to consider what intervals we get from the real line after deleting all these roots.



The intervals are $(-\infty, -3)$, $(-3, 0)$, $(0, 2)$, $(2, \infty)$.

Step 3): Next we make our sign table (and fill it in by picking up one test point from each interval):

	$(-\infty, -3)$	$(-3, 0)$	$(0, 2)$	$(2, \infty)$
x	$-$	$-$	$+$	$+$
$x - 2$	$-$	$-$	$-$	$+$
$x + 3$	$-$	$+$	$+$	$+$
$\frac{x(x-2)}{x+3}$	$-$	$+$	$-$	$+$

We see that the rational expression $\frac{x(x-2)}{x+3}$ is negative only on the intervals $(-\infty, -3)$ and $(0, 2)$. So our final answer is $(-\infty, -3) \cup (0, 2)$.

BONUS: We mentioned that for a quadratic equation, $ax^2 + bx + c = 0$, where a, b, c are real coefficients, there are only three possibilities for the roots of the equation: we may have two distinct real solutions, one unique real solution, or two distinct complex solutions.

But why can't we have just one unique non-real complex solution? e.g. why isn't there a quadratic equation with real coefficients whose only root is i ? Explain.

There are many ways to approach this problem. One is via the quadratic formula. The quadratic formula gives us the roots of any quadratic polynomial, $ax^2 + bx + c$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If our coefficients, a, b, c , are all real, then the only way we can get at least one non-real complex root is if $b^2 - 4ac$ be negative, so that our square-root is imaginary (since every other part of the formula, the $-b$, the $2a$ in the denominator is purely real). But then, supposing that $b^2 - 4ac$ **is** negative, it has some **nonzero** purely imaginary square root, say di where d is real, and then our solutions are:

$$x = \frac{-b \pm di}{2a} = \frac{-b}{2a} \pm \frac{d}{2a}i$$

and since all the numbers a, b, d are real and d is not zero, the roots $\frac{-b}{2a} + \frac{d}{2a}i$ and $\frac{-b}{2a} - \frac{d}{2a}i$ must be complex and distinct.